

Examining Heat Stress and Noise in Construction: A Reddit-Based Investigation of Workplace Challenges

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ABSTRACT

Construction workers face significant challenges from workplace stressors such as heat exposure and noise. This study investigates these issues by analyzing a construction-focused dataset mined from Reddit using social media data analysis techniques. A structured keyword-pairing strategy yielded 7,661 posts during the summer of 2024. Of these, 4,500 posts were manually reviewed, identifying 642 samples directly relevant to construction workers' health concerns. Advanced NLP techniques, such as BERTopic modeling, were employed to uncover key insights. The findings revealed that Reddit users actively discuss these environmental challenges, with Texas being the most frequently mentioned location. Using BERTopic modeling, key themes emerged, including “Heat, Hot, Construction, Water” and “Mental Health, Work, Health,” highlighting the environmental stressors and mental health concerns in construction.

INTRODUCTION

Construction workers face unique challenges in the work environment, such as noise and heat exposure, which could potentially lead to adverse mental health outcomes. Exposure to high levels of noise is strongly associated with increased psychological distress (Monazzam Esmaelpour, 2022; Scinicariello et al., 2019), in addition to permanent hearing loss, reduced

work productivity, and workplace accidents due to missing warning signals (CDC, 2023; OSHA, 2011). Further complicating the work environment, outdoor construction workers are vulnerable to heat stress, especially during summer when exposed to extreme temperatures.

Despite the recognized risks, there is limited evidence directly linking specific environmental factors to mental health issues among construction workers in the United States. Some studies, however, have highlighted the health impacts of extreme heat. For example, Karthick et al. (2022) reported that extreme hot weather conditions led to difficulties concentrating and frequent mood fluctuations based on responses from 100 surveyed workers. Shin et al. (2024) further demonstrated, using EEG metrics in immersive VR-based construction safety training, that extreme heat diminishes attention, increases fatigue, and heightens physiological arousal.

Given the stigma and barriers surrounding mental health discussions in traditional research settings, social media platforms like Reddit provide a unique opportunity to access insights from workers. These platforms enable researchers to capture authentic narratives and patterns that might otherwise go unnoticed. Additionally, Reddit offers a platform to identify other potential risk factors and environmental challenges based on workers' shared experiences, broadening the scope of understanding beyond what traditional methods might uncover.

This paper investigates the mental health implications of the work environment in construction, leveraging data mining of Reddit posts as a key methodology to explore these risks and identify additional factors impacting workers' mental health.

DATA COLLECTION

We initially identified a list of keywords with expert assistance to extract data from Reddit, which included 18 keywords: construction, pressure, burnout, depression, temperature, noise, heat, health, heatwave, hyperthermia, sunburn, irritation, dehydration, difficulties, stress, tension, mental, and fatigue. Data from all subreddits using a free Python package called PRAW (Python Reddit API Wrapper) were gathered. As of August 6, 2024, Reddit has around 138,000 active subreddits out of 1.2 million. PRAW uses the client ID, client secret, and user agent to authenticate with the Reddit API and obtain access tokens for authorized requests. These credentials identify the application and enable it to make API calls. With these tokens, PRAW interacts with the Reddit API endpoints to retrieve data, such as posts and comments, with a maximum of 1000 items per query. For each extracted post or comment, we collected the post or comment ID, user ID, text, date, subreddit, and upvotes. As Reddit does not provide location data for posts or comments through its API and users' location data is typically private, we excluded location data from this project.

We followed a structured strategy for searching posts or comments utilizing pairs of keywords with an 'AND' operator. We divided the list of keywords into two groups:

$$X = \{construction\}$$

$Y = \{pressure, burnout, depression, temperature, noise, heat, health, heatwave, hyperthermia, sunburn, irritation, dehydration, difficulties, stress, tension, mental, fatigue\}$

We tried all possible pairs (x, y) , where $x \in X$ and $y \in Y$. For example, we searched for pairs such as (construction, pressure), (construction, burnout), etc. The total number of keyword pairs was given by $|X| \times |Y|$, which is $1 \times 17 = 17$. To maintain the accuracy of the search, we converted each post or comment into lowercase before matching keyword pairs. We examined the subreddits in our initial collection and identified the most frequent subreddits related to construction. This list included Construction, ConstructionManagers, ConstructionTech, Plumbing, ConstructionBlog, ConstructionSupport, hvacadvice, civilengineering, HVAC, AusRenovation, Homebuilding, aviation, and flying. We then searched by all single keywords from set Y over these subreddits to gather more data. Finally, this strategy resulted in 7,661 posts or comments.

DATA CLEANING

Extracting data from social media using keywords often results in significant irrelevant data, particularly in Natural Language Processing (NLP) tasks. We found many irrelevant posts and comments that were not related to construction workers in our initial data. Identifying relevant posts and comments through manual review is a time-consuming task. We followed the process below to manually determine the relevance of each post and comment:

- First, we read the post or comment and identified whether it was related to construction. We primarily looked for some words, e.g., construction, concrete, building, plumbing, electrical, carpentry, HVAC, contractor, worksite, construction site, crane, etc. Note that the presence of such words does not guarantee that the post or comment was related to construction. For example, *“not sure what the claim is here since the usage of mental construction is vague. would all claims be a mental construction, including the claim about them being relative. how are you getting outside of the mental construction if we agree that all conceptual claims are "mental construction"?”* uses the word “construction” in a different context unrelated to building or construction work.
- If the post or comment was related to construction, we then determined if it discusses issues that cause difficulties for construction workers, potentially leading to stress, physical health issues, anxiety, or other challenges. We especially looked for the words, e.g., pressure, burnout, depression, noise, heat, health, heatwave, hyperthermia, sunburn, irritation, dehydration, difficulties, stress, tension, mental, fatigue, etc. Note that the presence of these words doesn’t guarantee that the post or

comment is related to construction work-related difficulties that can cause mental health problems.

Below are two relevant examples. For example, *“weird how people seem to think construction is an easy job where you just hit a nail with a hammer. its brutal on the body and people are very cruel. theres no real hr to help you when people are mean. you will fuck up a whole lot for the first few years and your. mental fortitude is probably not up to it. you can try. but dont go in thinking its like a ‘normal’ job where people have to act nice.”* is relevant to our task as it discusses the challenges faced by construction workers, highlighting issues such as physical strain, lack of support, and the impact on mental health. Another example, *“Making construction workers work in high temperatures and not letting them have a break, horrible,”* is relevant as it addresses the issue of heat stress and the lack of adequate breaks, which can contribute to mental health problems for construction workers. On the other hand, *“Construction noise central city overnight - Is anyone else experiencing a concrete drilling noise in the central city overnight? Im not sure what the laws are but seems a new townhouse complex it doing this loud work overnight.”* is not relevant to our task as it is about the inconvenience caused by construction noise to the general public, rather than faced by construction workers themselves. However, it was hard to identify some posts and comments as related to construction. For example, the following post, *“PROJECT MANAGERS!: What task do you find the most time-consuming and wish you could outsource? - I hear more and more about PM feeling burnout from work. They are losing time with their family and feel burnout frequently most likely do spending time on useless tasks that can be spent on more important tasks. ​ What do you guys think?”* doesn’t mention construction explicitly, but it discusses burnout from work. In such cases, we looked at the subreddit from which it was scrapped. If the subreddit is about construction (e.g., Construction, ConstructionManagers, ConstructionTech, Plumbing, etc.), we considered it relevant.

By applying the above approach, we have reviewed 4,500 out of the 7,661 samples from the construction dataset and identified 642 relevant samples. Using these 642 samples, we conducted a preliminary data analysis outlined in the following section.

DATA ANALYSIS AND RESULTS

Location analysis. Although the Reddit API does not provide information related to users' locations, people often mention places in their posts and comments. For example:

“What’s up with Texas and Florida not wanting outdoor workers to take breaks from the heat? - Texas passed legislation removing the requirement for farm and construction workers to have water and heat breaks. Florida just did the same and also blocked (locally) a [Miami-Dade effort to obtain an exception.] I’m admittedly not well versed on this topic, I just keep seeing the headlines. As someone who lives in Florida, this seems not just unfair but actually dangerous to the lives of those workers. It’s hot AF here already. What gives?.”

To understand the geographic distribution of mentions within the dataset, we utilized spaCy's English model “en_core_web_sm” for location extraction. This model is adept at recognizing named entities such as geopolitical locations (e.g., cities, states, countries) from text data. After processing the posts and comments, we extracted all location-related mentions and aggregated their frequencies.

We categorized each extracted location under its corresponding country. For instance, cities and states such as "Texas", "Florida", and "Oregon" were grouped under the United States (USA). Similarly, mentions like "Dubai" were attributed to the United Arab Emirates (UAE). The frequency of mentions for each country was calculated by summing up the occurrences of all associated locations. Figure 1 illustrates the frequency of the top five countries, showing that the USA had the highest frequency of mentions, with 64 occurrences, significantly surpassing other countries such as the UK (5 mentions) and the UAE (2 mentions).

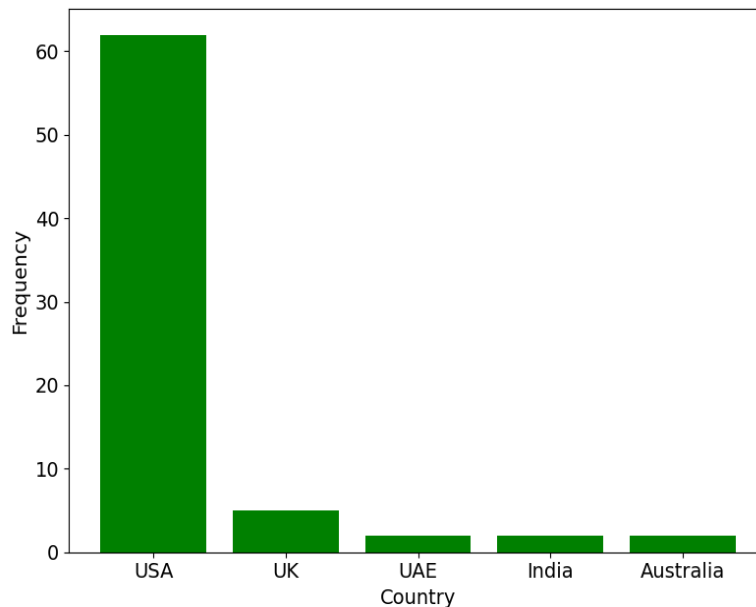


Figure 1. Distribution of the top 5 countries.

Given that the USA had the highest frequency of mentions, we delved deeper into the data to analyze which U.S. states were mentioned most frequently. Cities such as "Dallas" and "Boston" were mapped to their respective states (Texas and Massachusetts, respectively), and the state-level frequencies were calculated accordingly. Texas emerged as the most frequently mentioned state, with 32 mentions, followed by Florida (9 mentions) and Oregon (4 mentions). Figure 2 depicts the frequency distribution of the top 10 U.S. states. For this analysis, mentions of "US" and "USA" as country references were excluded to ensure state-level granularity.

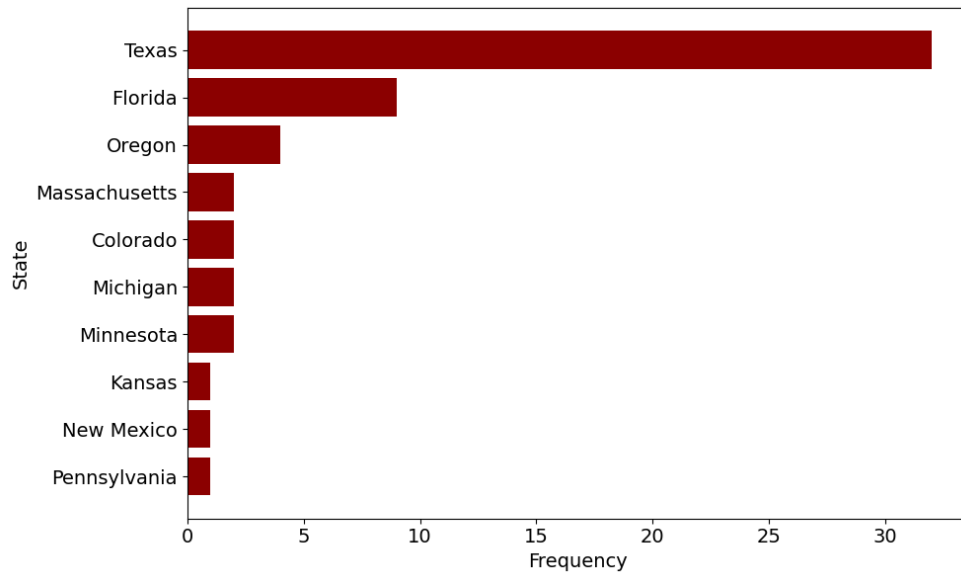


Figure 2. Frequency of the top 10 U.S. states

Subreddit. The dataset contains 642 posts and comments collected from 129 unique subreddits. Among these, the top five subreddits based on sample representation are ‘Construction,’ ‘antiwork,’ ‘ConstructionManagers,’ ‘politics,’ and ‘trolley problem.’ ‘Construction’ contributed the largest portion of the dataset, with 387 samples, followed by ‘antiwork’ with 35 samples and ‘ConstructionManagers’ with 32 samples. Subreddits like ‘politics’ and ‘trolleyproblem’ had smaller contributions, with 11 and 6 samples, respectively

Word cloud representation. The word cloud in Figure 3 visually summarizes the most frequently used words in the construction dataset. In this representation, the size of each word reflects its frequency, with larger words appearing more often than smaller ones. The prominence of the word ‘construction’ indicates that this term is central to the dataset, as expected. Additionally, words such as ‘work,’ ‘worker,’ ‘job,’ ‘day,’ ‘time,’ and ‘project’ highlight the focus on day-to-day tasks and projects within the construction industry. Terms like ‘mental health,’ ‘stress,’ ‘safety,’ ‘hot,’ ‘heat,’ ‘heat stroke,’ and ‘noise’ frequently appear, highlighting the challenges and difficulties construction workers encounter in their jobs.

dataset. Each topic is represented by its ID, the count of associated text samples, and a brief key theme derived from the most prominent keywords in the cluster with examples.

- **Topic 1** (Heat, Hot, Construction, Water): With 310 samples, this is the most prominent topic, reflecting discussions about the impact of heat and hydration on construction work. One example is *“since you mention the National Guard, i hope they’re getting breaks on the border! it was 130°F today in the sun. In Houston. hotter in the valley. construction guys should stay home and not risk it in the 'unprecedented' heat wave. Fuck Gregg Abbott. He’s an asshole”*
- **Topic 2** (Mental Health, Work, Health): The second largest topic (215 samples) focuses on mental health challenges and overall well-being within the construction industry. One example is *“mental health days are super important. stay tuned for the upcoming mental health awareness week for the construction industry next september during national suicide prevention month.”*
- **Topic 3** (Noise and Hearing Protection): This topic highlights concerns around noise levels and hearing safety, affecting 53 samples. One example is *“Even doubling up on hearing protection is not going to cut out all the noise. The point of these items are to protect your ears from damage, not cut sound out completely. You still need to be able to hear what is going on for safety reasons.”*
- **Topic 4** (Health and Safety): Discussions in this category (33 samples) revolve around worker safety and health-related practices. One example is *“Great point. I’m not in construction, but I lived in Texas and I always worried about construction safety during the hot months. Extreme heat has such an impact on judgment. Even the smartest person I know would start to make bad decisions after several hours of exposure to extreme heat. There needs to be at least more breaks or limited labor hours during the hottest months. I’d actually prefer major projects to go on hold during the hottest months.”*
- **Topic 5** (Masks and Dust): With 22 samples, this topic captures conversations about wearing protective masks and dealing with dust in construction environments. One example: *“Wear a mask every day to protect yourself from drywall dust, regular dust, wood dust is carcinogenic, paint, asbestos. The reason construction is bad for your health is because most people are idiots and don’t wear masks and breathe in these things on a daily basis. Also, protect your back, ears, and eyes.”*

Figure 4 represents the top 5 topics with the top 5 words based on c-TF-IDF scores, where the score indicates the relevance of each word to its respective topic. A higher score reflects greater relevance of the word to the topic, providing insight into the key themes and terms characterizing each topic. This visualization highlights the most influential words within each topic, aiding in understanding the primary focus areas of the dataset.

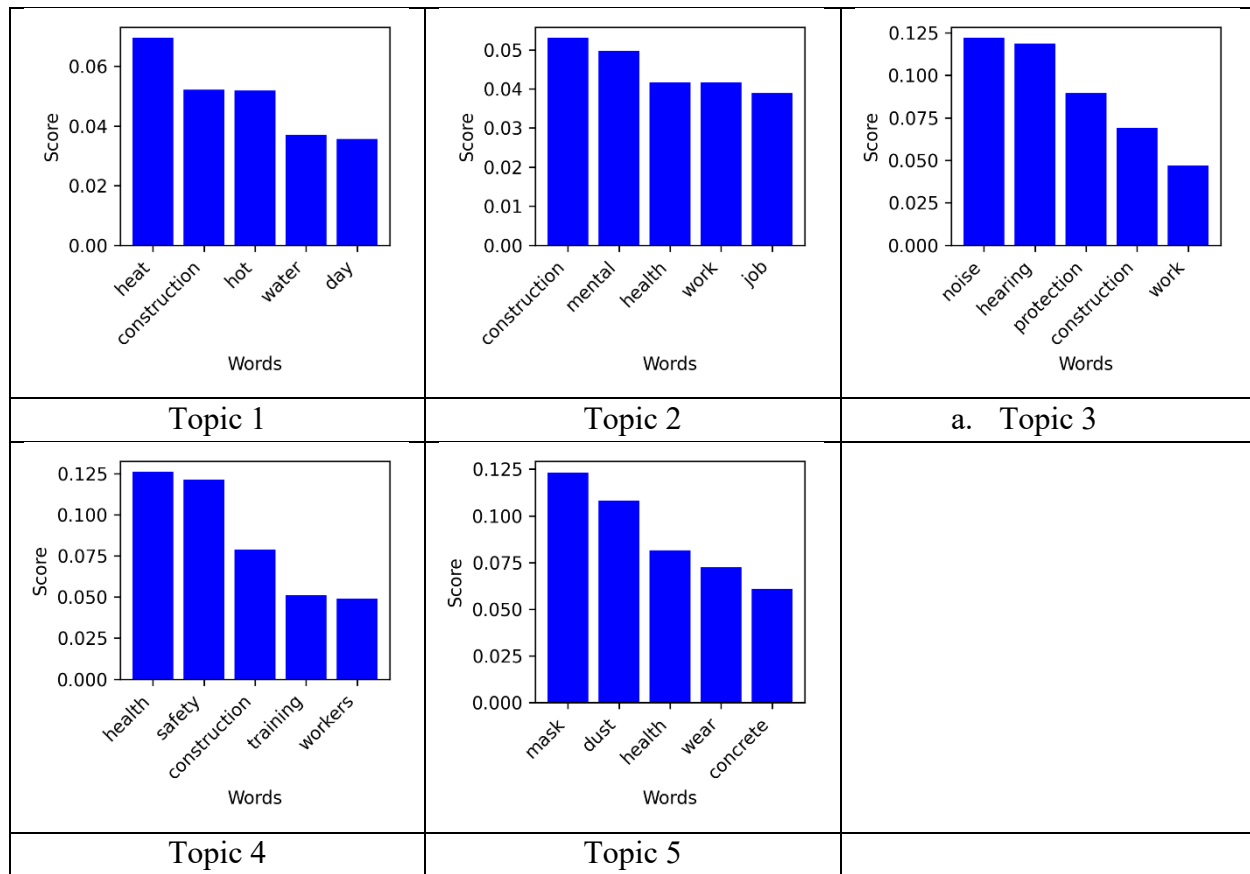


Figure 4. Top 5 words of the top 5 topics based on c-TF-IDF Scores.

DISCUSSION

This study highlights the potential of leveraging social media data to explore the mental health impacts of environmental stressors in the construction industry. Analyzing 642 relevant Reddit discussions demonstrates that heat stress and noise are recurring concerns among construction workers. The frequent mention of Texas underscores the region's vulnerability to extreme weather conditions, suggesting location-specific interventions are needed.

BERTopic modeling identified key themes linking environmental factors such as heat and noise to mental health and workplace challenges. These themes reflect the broader implications of occupational stressors on worker well-being, providing valuable insights into the lived experiences of construction workers. Furthermore, the study illustrates the utility of Reddit as a platform for uncovering authentic narratives and previously unreported issues, making it a promising tool for future research into occupational health risks.

CONCLUSION

This paper presents a preliminary analysis of a social media dataset from Reddit, examining the impacts of heat stress and noise in the construction industry. The authors reviewed 4,500 out of 7,661 posts/comments and identified 642 relevant discussions, forming the basis of the preliminary analysis. The findings reveal that Reddit users actively discuss these environmental challenges, with Texas being the most frequently mentioned location. Using BERTopic modeling, key themes emerged, including “Heat, Hot, Construction, Water” and “Mental Health, Work, Health,” highlighting the intersection of environmental stressors and mental health concerns in the construction sector.

ACKNOWLEDGMENT

A seed grant from the School of Data Science and the Charlotte AIR Institute at the University of North Carolina at Charlotte funded this project.

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